

矩阵分析与应用

第十一讲 矩阵分析及其应用之二

本讲主要内容

- 矩阵函数的一般定义
- 矩阵函数的性质
- 矩阵的微分和积分

三、矩阵函数的一般定义

展开式 $f(z) = \sum c_k z^k$, $(|z| < r, r > 0)$, 要求

$$(1) \quad f^{(k)}(0) \text{ 存在} \quad (k = 0, 1, 2, \dots)$$

$$(2) \quad \lim_{k \rightarrow \infty} \frac{f^{(k+1)}(\xi)}{(k+1)!} \xi^{k+1} = 0 \quad (|z| < r)$$

对于一元函数 $f(z) = \frac{1}{z}$ 等, 还不能定义矩阵函数。

基于矩阵函数值的**Jordan**标准形算法, 拓宽定义

矩阵函数的一般定义

设 $P^{-1}AP = J = \text{diag}(J_1, \dots, J_s)$, $J_i = \lambda_i I + I^{(1)}$

如果 $f(z)$ 在 λ_i 处有 $m_i - 1$ 阶导数, 令

$$f(J_i) = \sum_{k=0}^{\infty} c_k J_i^k = f(\lambda_i)I + \frac{f'(\lambda_i)}{1!} I^{(1)} + \dots + \frac{f^{(m_i-1)}(\lambda_i)}{(m_i-1)!} I^{(m_i-1)}$$

$$f(A) = \sum_{k=0}^{\infty} c_k A^k = P \cdot \sum_{k=0}^{\infty} c_k J^k \cdot P^{-1} = P \cdot \text{diag}(f(J_1), \dots, f(J_s)) \cdot P^{-1}$$

称 $f(A)$ 为对应于 $f(z)$ 的矩阵函数

[注] ① 拓宽定义不要求 $f(z)$ 能展为 “ z ” 的幂级数，
但要求在 A 的特征值 λ_i （重数为 m_i ）处有
 $m_i - 1$ 阶导数，后者较前者弱！

② 当能够展为 “ z ” 的幂级数时，矩阵函数的拓宽
定义与级数原始定义是一致的。

例9: $A = \begin{bmatrix} 2 & 1 & & \\ & 2 & 1 & \\ & & 2 & 1 \\ & & & 2 \end{bmatrix}$, $f(z) = \frac{1}{z}$, 求 $f(A)$

解: $f(z) = \frac{1}{z}$, $f'(z) = -z^{-2}$, $f''(z) = 2z^{-3}$, $f'''(z) = -6z^{-4}$

$$f(A) = f(J)$$

$$= f(2) \cdot I + f'(2) \cdot I^{(1)} + \frac{f''(2)}{2!} \cdot I^{(2)} + \frac{f'''(2)}{3!} \cdot I^{(3)}$$

$$= \begin{bmatrix} 0.5 & -0.25 & 0.125 & -0.0625 \\ & 0.5 & -0.25 & 0.125 \\ & & 0.5 & -0.25 \\ & & & 0.5 \end{bmatrix}$$

例10: $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, $f(z) = \sqrt{z}$, 求 $f(A)$

解: $f(z) = \sqrt{z}$, $f'(z) = \frac{1}{2\sqrt{z}}$

$$J_1 = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}: f(J_1) = f(1) \cdot I + f'(1) \cdot I^{(1)} = \begin{bmatrix} 1 & 1/2 \\ & 1 \end{bmatrix}$$

$$J_2 = [2]: f(J_2) = f(2) \cdot I = [\sqrt{2}]$$

$$f(A) = f(J) = \begin{bmatrix} f(J_1) & & \\ & f(J_2) & \\ & & \end{bmatrix} = \begin{bmatrix} 1 & 1/2 & 0 \\ & 1 & 0 \\ & & \sqrt{2} \end{bmatrix}$$

四、矩阵函数的性质

级数定义或拓宽定义给出的矩阵函数具有下列性质：

$$(1) \quad f(z) = f_1(z) + f_2(z) \Rightarrow f(A) = f_1(A) + f_2(A)$$

$$f^{(l)}(\lambda_i) = f_1^{(l)}(\lambda_i) + f_2^{(l)}(\lambda_i)$$

$$\Rightarrow f^{(l)}(J_i) = f_1^{(l)}(J_i) + f_2^{(l)}(J_i)$$

$$\begin{aligned} f(A) &= P \bullet \left\{ \begin{bmatrix} f_1(J_1) & & \\ & \ddots & \\ & & f_1(J_s) \end{bmatrix} + \begin{bmatrix} f_2(J_1) & & \\ & \ddots & \\ & & f_2(J_s) \end{bmatrix} \right\} \bullet P^{-1} \\ &= f_1(A) + f_2(A) \end{aligned}$$

$$(2) \quad f(z) = f_1(z) \bullet f_2(z)$$

$$\Rightarrow f(A) = f_1(A) \bullet f_2(A) = f_2(A) \bullet f_1(A)$$

$$\begin{aligned} f_1(J_i) \bullet f_2(J_i) &= \left[f_1 \bullet I + f_1' \bullet I^{(1)} + \frac{f_1''}{2!} \bullet I^{(2)} + \dots + \frac{f_1^{(m_i-1)}}{(m_i-1)!} \bullet I^{(m_i-1)} \right] \bullet \\ &\quad \left[f_2 \bullet I + \frac{f_2'}{1!} \bullet I^{(1)} + \frac{f_2''}{2!} \bullet I^{(2)} + \dots + \frac{f_2^{(m_i-1)}}{(m_i-1)!} \bullet I^{(m_i-1)} \right] \\ &= (f_1 f_2) \bullet I + \frac{f_1' f_2 + f_1 f_2'}{1!} \bullet I^{(1)} + \frac{f_1'' f_2 + 2 f_1' f_2' + f_1 f_2''}{2!} \bullet I^{(2)} + \dots \\ &= (f_1 f_2) \bullet I + \frac{(f_1 f_2)'}{1!} \bullet I^{(1)} + \frac{(f_1 f_2)''}{2!} \bullet I^{(2)} + \dots + \frac{(f_1 f_2)^{(m_i-1)}}{(m_i-1)!} \bullet I^{(m_i-1)} \\ &= f(J_i) \end{aligned}$$

$$f(A) = P \begin{bmatrix} f(J_1) & & \\ & \ddots & \\ & & f(J_s) \end{bmatrix} P^{-1}$$

$$= P \begin{bmatrix} f_1(J_1) & & \\ & \ddots & \\ & & f_1(J_s) \end{bmatrix} P^{-1} \bullet P \begin{bmatrix} f_2(J_1) & & \\ & \ddots & \\ & & f_2(J_s) \end{bmatrix} P^{-1}$$

$$= f_1(A) \bullet f_2(A)$$

4、矩阵的微分和积分

定义: 如果矩阵 $A(t) = (a_{ij}(t))_{m \times n}$, 的每一个元素 $a_{ij}(t)$ 是变量 t 的可微函数, 则 $A(t)$ 关于 t 的导数(微商)定义为

$$\frac{dA(t)}{dt} = (a'_{ij}(t))_{m \times n}, \text{ 或者 } A'(t) = (a'_{ij}(t))_{m \times n}$$

定理8: 设 $A(t), B(t)$ 可导, 则有

$$(1) \quad \frac{d}{dt}[A(t) + B(t)] = \frac{d}{dt}A(t) + \frac{d}{dt}B(t)$$

$$(2) \quad A_{m \times n}, f(t) \text{ 可导 } \frac{d}{dt}[f(t)A(t)] = f'(t)A(t) + f(t)A'(t)$$

$$(3) \quad A_{m \times n}, B_{n \times l} : \frac{d}{dt}[A(t)B(t)] = A'(t)B(t) + A(t)B'(t)$$

$$\begin{aligned} \text{证明: } (3) \text{ 左} &= \frac{d}{dt} \left(\sum_k a_{ik}(t) b_{kj}(t) \right)_{m \times l} \\ &= \left(\sum_k a'_{ik}(t) b_{kj}(t) + \sum_k a_{ik}(t) b'_{kj}(t) \right)_{m \times l} \\ &= \left(\sum_k a'_{ik}(t) b_{kj}(t) \right)_{m \times l} + \left(\sum_k a_{ik}(t) b'_{kj}(t) \right)_{m \times l} = \text{右} \end{aligned}$$

定理9: 设 $A_{n \times n}$ 为常数矩阵, 则有

$$(1) \quad \frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A$$

$$(2) \quad \frac{d}{dt} \cos(tA) = -A \bullet \sin(tA) = -\sin(tA) \bullet A$$

$$(3) \quad \frac{d}{dt} \sin(tA) = A \bullet \cos(tA) = \cos(tA) \bullet A$$

证明: (1) $e^{tA} = \sum_{n=0}^{\infty} \frac{1}{n!} (tA)^n$ 绝对收敛

$$(e^{tA})_{ij} = \delta_{ij} + \frac{t}{1!} (A)_{ij} + \frac{t^2}{2!} (A^2)_{ij} + \cdots + \frac{t^k}{k!} (A^k)_{ij} + \cdots \text{ 绝对收敛}$$

$$\left(e^{tA}\right)_{ij} = \delta_{ij} + \frac{t}{1!}(A)_{ij} + \frac{t^2}{2!}(A^2)_{ij} + \cdots + \frac{t^k}{k!}(A^k)_{ij} + \cdots \quad \text{绝对收敛}$$

$$\frac{d}{dt}\left(e^{tA}\right)_{ij} = 0 + (A)_{ij} + \frac{t}{1!}(A^2)_{ij} + \cdots + \frac{t^{k-1}}{(k-1)!}(A^k)_{ij} + \cdots$$

绝对收敛

$$\frac{d}{dt}e^{tA} = A + \frac{t}{1!}A^2 + \cdots + \frac{t^{k-1}}{(k-1)!}A^k + \cdots$$

绝对收敛

$$= \begin{cases} A \left[I + \frac{t}{1!}A + \cdots + \frac{t^{k-1}}{(k-1)!}A^{k-1} + \cdots \right] & = Ae^{tA} \\ \left[I + \frac{t}{1!}A + \cdots + \frac{t^{k-1}}{(k-1)!}A^{k-1} + \cdots \right] A & = e^{tA}A \end{cases}$$

例11: $A(t) = (a_{ij}(t))_{n \times n}$ 可逆, 则

$$\frac{dA^{-1}(t)}{dt} = -A^{-1}(t) \frac{dA(t)}{dt} A^{-1}(t)$$

解: 对于恒等式 $A^{-1}(t)A(t) = I$

两边求导
$$\frac{dA^{-1}(t)}{dt} A(t) + A^{-1}(t) \frac{dA(t)}{dt} = 0$$

可得
$$\frac{dA^{-1}(t)}{dt} = -A^{-1}(t) \frac{dA(t)}{dt} A^{-1}(t)$$

定义: 如果矩阵 $A(t) = (a_{ij}(t))_{m \times n}$ 的每一个元素 $a_{ij}(t)$

在 $[t_0, t]$ 上可积, 称 $A(t)$ 可积, 记为

$$\int_{t_0}^t A(\tau) d\tau = \left(\int_{t_0}^t a_{ij}(\tau) d\tau \right)_{m \times n}$$

$$(1) \int_{t_0}^t [A(\tau) + B(\tau)] d\tau = \int_{t_0}^t A(\tau) d\tau + \int_{t_0}^t B(\tau) d\tau$$

$$(2) A \text{ 为常数矩阵: } \int_{t_0}^t [A \bullet B(\tau)] d\tau = A \bullet \left[\int_{t_0}^t B(\tau) d\tau \right]$$

$$B \text{ 为常数矩阵: } \int_{t_0}^t [A(\tau) \bullet B] d\tau = \left[\int_{t_0}^t A(\tau) d\tau \right] \bullet B$$

$$(3) \text{ 设 } a_{ij}(t) \in C[t_0, t_1] \quad \text{则: } \frac{d}{dt} \int_{t_0}^t A(\tau) d\tau = A(t)$$

$$(4) \text{ 设 } a'_{ij}(t) \in C[t_0, t_1], \quad \text{则: } \int_{t_0}^{t_1} A'(\tau) d\tau = A(t_1) - A(t_0)$$

其它微分概念

函数对矩阵的导数(包括向量)

定义: 设 $X = (\xi_{ij})_{m \times n}$, mn 元函数

$$f(X) = f(\xi_{11}, \xi_{12}, \dots, \xi_{1n}, \dots, \xi_{m \times n})$$

定义 $f(X)$ 对矩阵 X 的导数为

$$\frac{df}{dX} = \left(\frac{\partial f}{\partial \xi_{ij}} \right)_{m \times n} = \begin{bmatrix} \frac{\partial f}{\partial \xi_{11}} & \dots & \frac{\partial f}{\partial \xi_{1n}} \\ \vdots & & \vdots \\ \frac{\partial f}{\partial \xi_{m1}} & \dots & \frac{\partial f}{\partial \xi_{mn}} \end{bmatrix}$$

例12: $\mathbf{x} = \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix} : f(\mathbf{x}) = f(\xi_1, \xi_2, \dots, \xi_n) \quad \frac{df}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f}{\partial \xi_1} \\ \vdots \\ \frac{\partial f}{\partial \xi_n} \end{bmatrix}$

例13: $A = (a_{ij})_{n \times n}, \mathbf{x} = \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix} : f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}, \text{ 求 } \frac{df}{d\mathbf{x}}$

$$f(\mathbf{x}) = \xi_1 \sum_{j=1}^n a_{1j} \xi_j + \dots + \xi_k \sum_{j=1}^n a_{kj} \xi_j + \dots + \xi_n \sum_{j=1}^n a_{nj} \xi_j$$

$$\begin{aligned} \frac{\partial f}{\partial \xi_k} &= \xi_1 a_{1k} + \dots + \xi_{k-1} a_{k-1,k} + \left(\sum_{j=1}^n a_{kj} \xi_j + \xi_k a_{kk} \right) \\ &\quad + \xi_{k+1} a_{k+1,k} + \dots + \xi_n a_{nk} = \sum_{j=1}^n a_{kj} \xi_j + \sum_{i=1}^n a_{ik} \xi_i \end{aligned}$$

$$\therefore \frac{df}{dx} = (A + A^T)x$$

如果 $A = A^T$, 有 $\frac{df}{dx} = 2Ax$

例14: $X = (\xi_{ij})_{n \times n} : f(X) = [\text{tr}(X)]^2$ 求 $\left. \frac{df}{dX} \right|_{X=I_n}$

解: $f(X) = (\xi_{11} + \xi_{22} + \cdots + \xi_{nn})^2$

$$\frac{df}{dX} = 2(\xi_{11} + \xi_{22} + \cdots + \xi_{nn}) I_n$$

$$\left. \frac{df}{dX} \right|_{X=I_n} = 2nI_n$$

例15: $A \in R^{m \times n}, b \in R^n$, 若 $x \in R^n$ 使得 $\|Ax - b\|_2 = \min$,

则 $A^T Ax = A^T b$

解: $f(x) = \|Ax - b\|_2^2 = (Ax - b)^T (Ax - b)$

$$= x^T A^T Ax - 2b^T Ax + b^T b$$

$$g(x) = b^T Ax = b_1 \sum_{j=1}^n a_{1j} \xi_j + \cdots + b_m \sum_{j=1}^n a_{mj} \xi_j$$

$$\frac{dg}{dx} = \begin{bmatrix} \frac{\partial g}{\partial \xi_1} \\ \vdots \\ \frac{\partial g}{\partial \xi_n} \end{bmatrix} = \begin{bmatrix} b_1 a_{11} + \cdots + b_m a_{m1} \\ \vdots \\ b_1 a_{1n} + \cdots + b_m a_{mn} \end{bmatrix} = A^T b$$

$$\frac{df}{dx} = 2A^T Ax - 2A^T b = 0 \Rightarrow A^T Ax = A^T b$$

【注】 $r(A^T A) = r(A) \Rightarrow r(A^T A | A^T b) = r(A^T A) \Rightarrow A^T Ax = A^T b$ 有解

5、函数矩阵对矩阵的导数

定义: 设 $X = (\xi_{ij})_{m \times n}$, $f_{kl}(X) = f_{kl}(\xi_{11}, \xi_{12}, \dots, \xi_{1n}, \dots, \xi_{m \times n})$

$$F = \begin{bmatrix} f_{11} & \cdots & f_{1s} \\ \vdots & & \vdots \\ f_{r1} & \cdots & f_{rs} \end{bmatrix}, \quad \frac{\partial F}{\partial \xi_{ij}} = \begin{bmatrix} \frac{\partial f_{11}}{\partial \xi_{ij}} & \cdots & \frac{\partial f_{1s}}{\partial \xi_{ij}} \\ \vdots & & \vdots \\ \frac{\partial f_{r1}}{\partial \xi_{ij}} & \cdots & \frac{\partial f_{rs}}{\partial \xi_{ij}} \end{bmatrix},$$

$$\text{定义 } \frac{dF}{dX} = \begin{bmatrix} \frac{\partial F}{\partial \xi_{11}} & \cdots & \frac{\partial F}{\partial \xi_{1n}} \\ \vdots & & \vdots \\ \frac{\partial F}{\partial \xi_{m1}} & \cdots & \frac{\partial F}{\partial \xi_{mn}} \end{bmatrix}$$

■ 可表示为

$$\frac{dF}{dX} = \left(\frac{1}{dX} \right) \otimes dF$$

例16: $\mathbf{x} = \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix}$, $F(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_l(\mathbf{x})]$

$$\frac{dF}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial \xi_1} & \dots & \frac{\partial f_l}{\partial \xi_1} \\ \vdots & & \vdots \\ \frac{\partial f_1}{\partial \xi_n} & \dots & \frac{\partial f_l}{\partial \xi_n} \end{bmatrix}$$

例17: $A = (a_{ij})_{n \times n}$, $\mathbf{x} = \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix}$ $A\mathbf{x} = \begin{bmatrix} \sum_{j=1}^n a_{1j} \xi_j \\ \vdots \\ \sum_{j=1}^n a_{nj} \xi_j \end{bmatrix}$

$$\frac{d(A\mathbf{x})}{d\mathbf{x}^T} = \begin{bmatrix} \frac{d(A\mathbf{x})}{d\xi_1} & \dots & \frac{d(A\mathbf{x})}{d\xi_n} \end{bmatrix} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} = A$$

定理10（转换定理）：设 $X_{m \times n} = [x_{ij}]$, $Y_{p \times q} = [y_{ij}]$
 矩阵 $A_{p \times m}$, $B_{n \times q}$, $C_{p \times n}$, $D_{m \times q}$ 是 X 的函数，则下列两条等价

$$(1) \quad \frac{\partial Y}{\partial x_{ij}} = AE_{ij}B + CE_{ij}^T D, \quad i = 1, \dots, m; j = 1, \dots, n; E_{ij} \in R^{m \times n}$$

$$(2) \quad \frac{dy_{ij}}{dX} = A^T E_{ij} B^T + D E_{ij}^T C, \quad i = 1, \dots, p; j = 1, \dots, q; E_{ij} \in R^{p \times q}$$

证明： $e_i = (0, \dots, 1, \dots, 0)^T$ ，则 $E_{ij} = e_i e_j'$ ，维数依上下文确定

$$\begin{aligned} & e_k' (AE_{ij}B + CE_{ij}^T D) e_l = e_k' A e_i e_j' B e_l + e_k' C e_j e_i' D e_l \\ & = e_i' A' e_k e_l' B' e_j + e_i' D e_l e_k' C e_j = e_i' (A' e_k e_l' B' + D e_l e_k' C) e_j \\ & = e_i' (A' E_{kl} B' + D E_{lk} C) e_j \end{aligned}$$

$$\mathbf{e}'_k \left(\mathbf{A} \mathbf{E}_{ij} \mathbf{B} + \mathbf{C} \mathbf{E}_{ij}^T \mathbf{D} \right) \mathbf{e}_l = \mathbf{e}'_i \left(\mathbf{A}' \mathbf{E}_{kl} \mathbf{B}' + \mathbf{D} \mathbf{E}_{lk} \mathbf{C} \right) \mathbf{e}_j$$

如果 (1) 成立, 则矩阵 $\frac{\partial Y}{\partial \mathbf{x}_{ij}}$ 的 (k, l) 元

$$\left(\frac{\partial Y}{\partial \mathbf{x}_{ij}} \right)_{kl} = \mathbf{e}'_k \left(\mathbf{A} \mathbf{E}_{ij} \mathbf{B} + \mathbf{C} \mathbf{E}_{ij}^T \mathbf{D} \right) \mathbf{e}_l = \mathbf{e}'_i \left(\mathbf{A}' \mathbf{E}_{kl} \mathbf{B}' + \mathbf{D} \mathbf{E}_{lk} \mathbf{C} \right) \mathbf{e}_j$$

另外
$$\left(\frac{\partial Y}{\partial \mathbf{x}_{ij}} \right)_{kl} = \frac{\partial y_{kl}}{\partial \mathbf{x}_{ij}} = \left(\frac{\partial y_{kl}}{\partial \mathbf{X}} \right)_{ij} = \mathbf{e}'_i \left(\mathbf{A}' \mathbf{E}_{kl} \mathbf{B}' + \mathbf{D} \mathbf{E}_{lk} \mathbf{C} \right) \mathbf{e}_j$$

所以
$$\frac{\partial y_{ij}}{\partial \mathbf{X}} = \mathbf{A}' \mathbf{E}_{ij} \mathbf{B}' + \mathbf{D} \mathbf{E}'_{ij} \mathbf{C}$$

例18: $A \in R^{m \times n}$, $X \in R^{n \times m}$, 则 $\frac{d \operatorname{tr}(AX)}{dX} = A^T$

解: $\frac{d \operatorname{tr}(AX)}{dX} = \frac{d}{dX} \sum_{i=1}^m (AX)_{ii} = \sum_{i=1}^m \frac{\partial (AX)_{ii}}{\partial X}$

因为 $\frac{\partial AX}{\partial x_{ij}} = AE_{ij}$, $E_{ij} \in R^{n \times m}$ $\frac{\partial Y}{\partial x_{ij}} = AE_{ij}B + CE_{ij}^T D$

由转换定理 $\frac{\partial (AX)_{ij}}{\partial X} = A^T E_{ij}$, $E_{ij} \in R^{n \times n}$ $\frac{\partial y_{ij}}{\partial x} = A^T E_{ij} B^T + D E_{ij}^T C$

$$\frac{d \operatorname{tr}(AX)}{dX} = \sum_{i=1}^m A^T E_{ii} = A^T$$

【注】 $A \in R^{m \times n}$, $x \in R^n$, 则 $\frac{d \operatorname{tr}(Ax)}{dx} = A^T$

例19: $A_{m \times n}, X_{n \times k}, B_{k \times m}$ 则 $\frac{d \operatorname{tr}(AXB)}{dX} = A^T B^T$

解: $\frac{d \operatorname{tr}(AXB)}{dX} = \frac{d}{dX} \sum_{i=1}^m (AXB)_{ii} = \sum_{i=1}^m \frac{\partial (AXB)_{ii}}{\partial X}$

因为 $\frac{\partial AXB}{\partial x_{ij}} = AE_{ij}B, \quad E_{ij} \in R^{n \times k}$ $\frac{\partial Y}{\partial x_{ij}} = AE_{ij}B + CE_{ij}^T D$
 $\frac{dy_{ij}}{dX} = A^T E_{ij} B^T + DE_{ij}^T C$

由转换定理 $\frac{\partial (AXB)_{ij}}{\partial X} = A^T E_{ij} B^T, \quad E_{ij} \in R^{m \times m}$

$$\frac{d \operatorname{tr}(AXB)}{dX} = \sum_{i=1}^n A^T E_{ii} B^T = A^T B^T$$

例20: (1) $A_{n \times n}, X_{n \times m}$ 则 $\frac{d \operatorname{tr}(X'AX)}{dX} = (A + A^T)X$

(2) $A_{m \times m}, X_{n \times m}$ 则 $\frac{d \operatorname{tr}(XAX')}{dX} = X(A + A^T)$

解: $\frac{d \operatorname{tr}(X'AX)}{dX} = \sum_{i=1}^m \frac{\partial (X'AX)_{ii}}{\partial X}$

$$\frac{\partial Y}{\partial x_{ij}} = AE_{ij}B + CE_{ij}^T D$$

$$\frac{dy_{ij}}{dX} = A^T E_{ij} B^T + D E_{ij}^T C$$

因为 $\frac{\partial X'AX}{\partial x_{ij}} = E'_{ij}AX + X'AE_{ij} = \mathbf{X'A}E_{ij}I + IE'_{ij}\mathbf{AX}$

由转换定理 $\frac{\partial (X'AX)_{ii}}{\partial X} = A^T XE_{ij} + AX E_{ij}^T$

$$\frac{d \operatorname{tr}(X'AX)}{dX} = \sum_{i=1}^m (A^T XE_{ii} + AX E_{ii}^T) = (A + A^T)X$$

矩阵分析的应用

$$\begin{cases} \xi_1'(t) = a_{11}\xi_1(t) + a_{12}\xi_2(t) + \cdots + a_{1n}\xi_n(t) + b_1(t) \\ \xi_2'(t) = a_{21}\xi_1(t) + a_{22}\xi_2(t) + \cdots + a_{2n}\xi_n(t) + b_2(t) \\ \vdots \\ \xi_n'(t) = a_{n1}\xi_1(t) + a_{n2}\xi_2(t) + \cdots + a_{nn}\xi_n(t) + b_n(t) \end{cases}$$

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}, x(t) = \begin{bmatrix} \xi_1(t) \\ \vdots \\ \xi_n(t) \end{bmatrix}, b(t) = \begin{bmatrix} b_1(t) \\ \vdots \\ b_n(t) \end{bmatrix}, c(t) = \begin{bmatrix} c_1(t) \\ \vdots \\ c_n(t) \end{bmatrix}$$

齐次微分方程: $x'(t) = A \bullet x(t)$

非齐次微分方程: $x'(t) = A \bullet x(t) + b(t)$

齐次微分方程的解法

定理10: 齐次方程 $x'(t) = A \bullet x(t)$ 满足 $x(t_0) = x_0$ 的解存在并且唯一

证: 存在性 设 $x(t) = e^{(t-t_0)A} x_0$, 则

$$x'(t) = A e^{(t-t_0)A} x_0 = A \bullet x(t) \quad x(t_0) = e^0 x_0 = x_0$$

唯一性 设 $x(t)$ 满足 $x'(t) = A \bullet x(t), x(t_0) = x_0$

$$x'(t) - Ax(t) = 0 \Rightarrow e^{-tA} x'(t) + e^{-tA} (-A)x(t) = 0$$

$$\Rightarrow \left[e^{-tA} x(t) \right]' = 0 \Rightarrow e^{-tA} x(t) = c \Rightarrow x(t) = e^{tA} c$$

因为 $x(t_0) = x_0$, 所以 $x_0 = e^{t_0 A} c \Rightarrow c = e^{-t_0 A} x_0$

因此 $x(t) = e^{tA} e^{-t_0 A} x_0 = e^{(t-t_0)A} x_0$

例21: 设 $A = \begin{bmatrix} 1 & 1 & 0 \\ & 1 & 0 \\ & & 2 \end{bmatrix}$ 求 $x'(t) = A \cdot x(t)$ 的通解

解:
$$e^{tA} = \begin{bmatrix} e^t & te^t & 0 \\ & e^t & 0 \\ & & e^{2t} \end{bmatrix}$$

$$x(t) = e^{tA}c = \begin{bmatrix} c_1e^t + c_2te^t \\ c_2e^t \\ c_3e^{2t} \end{bmatrix}$$

例22: 矩阵函数 e^{tA} 的列向量 $x_1(t), \dots, x_n(t)$ 构成
齐次方程 $x'(t) = A \bullet x(t)$ 的基础解系

解: e^{tA} 可逆 $\Rightarrow x_1(t), \dots, x_n(t)$ 线性无关

取 $c = e_j \Rightarrow x_j(t) = e^{tA}c$ 是 $x'(t) = A \bullet x(t)$ 的一个解

通解 $x(t) = e^{tA}c = c_1 \bullet x_1(t) + \dots + c_n \bullet x_n(t)$

非齐次微分方程的解法

方程(1): $x'(t) = A \bullet x(t)$

方程(2): $x'(t) = A \bullet x(t) + b(t)$

$$\left. \begin{array}{l} \tilde{x}(t) \text{ 是(2)的特解} \\ x(t) \text{ 是(2)的通解} \end{array} \right\} \xrightarrow{\quad} \left\{ \begin{array}{l} \tilde{x}'(t) = A \bullet \tilde{x}(t) + b(t) \\ x'(t) = A \bullet x(t) + b(t) \end{array} \right.$$

$$\Rightarrow [x(t) - \tilde{x}(t)]' = A[x(t) - \tilde{x}(t)] \Rightarrow x(t) - \tilde{x}(t) \text{ 是(1)的解}$$

$$\Rightarrow x(t) - \tilde{x}(t) = c_1 \bullet x_1(t) + \cdots + c_n \bullet x_n(t)$$

$$\Rightarrow x(t) = e^{tA}c + \tilde{x}(t)$$

非齐次微分方程的解法

采用常向量变易法求 $\tilde{x}(t)$. 设 $\tilde{x}(t) = e^{tA}c(t)$ 满足(2), 有

$$Ae^{tA}c(t) + e^{tA}c'(t) = Ae^{tA}c(t) + b(t)$$

$$c'(t) = e^{-tA}b(t) \Rightarrow c(t) = \int_{t_0}^t e^{-\tau A}b(\tau)d\tau \quad (\text{原函数之一})$$

$$\text{故(2)的通解为 } x(t) = e^{tA} \left[c + \int_{t_0}^t e^{-\tau A}b(\tau)d\tau \right]$$

$$\text{特解为 } x(t)|_{x(t_0)=x_0} = e^{tA} \left[e^{-t_0A}x_0 + \int_{t_0}^t e^{-\tau A}b(\tau)d\tau \right]$$

$$[\text{注}] \text{ 当 } t_0 = 0 \text{ 时, 特解 } x(t)|_{x(0)=x_0} = e^{tA} \left[x_0 + \int_0^t e^{-\tau A}b(\tau)d\tau \right]$$

例23: 设 $A = \begin{bmatrix} 1 & 1 & 0 \\ & 1 & 0 \\ & & 2 \end{bmatrix}$, $b(t) = \begin{bmatrix} 1 \\ 0 \\ e^{2t} \end{bmatrix}$, $x(0) = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

求 $x'(t) = A \cdot x(t) + b(t)$ 满足初始条件 $x(0)$ 的特解

解: $e^{tA} = \begin{bmatrix} e^t & te^t & 0 \\ & e^t & 0 \\ & & e^{2t} \end{bmatrix}$

$$e^{-\tau A} b(\tau) = \begin{bmatrix} e^{-\tau} \\ 0 \\ 1 \end{bmatrix}, \quad \int_0^t e^{-\tau A} b(\tau) d\tau = \begin{bmatrix} 1 - e^{-t} \\ 0 \\ t \end{bmatrix}$$

$$x(t) = e^{tA} \cdot \left[\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 - e^{-t} \\ 0 \\ t \end{pmatrix} \right] = \begin{bmatrix} te^t - 1 \\ e^t \\ te^{2t} \end{bmatrix}$$

矩阵微分与最优化

最简单的最优化问题是求 $f(\mathbf{x})$ 的极大值和极小值

$$\min_{\mathbf{x} \in R} f(\mathbf{x})$$

一般称为无约束的最优化问题

相对于 $n \times 1$ 向量 $\mathbf{x}$ 的梯度算子记作 $\nabla_{\mathbf{x}}$

$$\nabla_{\mathbf{x}} = \left[\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \dots, \frac{\partial}{\partial x_n} \right]^T = \frac{\partial}{\partial \mathbf{x}}$$

$n \times 1$ 实向量 $\mathbf{x}$ 为变元的实标量函数的梯度

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \left[\frac{\partial f(\mathbf{x})}{\partial x_1}, \frac{\partial f(\mathbf{x})}{\partial x_2}, \dots, \frac{\partial f(\mathbf{x})}{\partial x_n} \right]^T = \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}}$$

矩阵微分与最优化

实标量函数 $f(\mathbf{A})$ 为相对于实矩阵 $\mathbf{A} = [\mathbf{a}_{ij}]_{m \times n}$ 的梯度

$$\frac{\partial f(\mathbf{A})}{\partial \mathbf{A}} = \left[\frac{\partial f(\mathbf{A})}{\partial \mathbf{a}_{ij}} \right]_{m \times n} = \nabla_{\mathbf{A}} f(\mathbf{A})$$

例：CDMA系统中，有 K 个用户，第 k 个用户的扩频波形向量为 $\mathbf{s}_k(t)$ 。假定用户 k 的信号幅值为 A_k

在 t 时刻发送比特为 b_k $(+1, -1)$

在基站解扩后，基站的接收信号向量为

$$\mathbf{y} = \mathbf{R}\mathbf{A}\mathbf{b} + \mathbf{n}$$

其中 $\mathbf{A} = \text{diag}(A_1, A_2, \dots, A_K)$, $\mathbf{b} = [b_1, b_2, \dots, b_K]^T$

扩频相关矩阵 $\mathbf{R}$ 的元素 $r_{ij} = \int_0^T s_i(t)s_j(t)dt$

设计一个多用户检测器 $\mathbf{M} = [m_1, m_2, \dots, m_K]$ ，使得

$$\hat{b}_k = \text{sgn}(m_k^T \mathbf{y})$$

将 K 个用户的检测器联合考虑，构造目标函数

$$J(\mathbf{M}) = \mathbf{E} \left[\|\mathbf{b} - \mathbf{M}\mathbf{y}\|_2^2 \right]$$

使其最小化，即可得到最优的盲多用户检测器 $\mathbf{M}$
利用矩阵迹的性质，可得

$$\begin{aligned} J(\mathbf{M}) &= \mathbf{E} \left\{ (\mathbf{b} - \mathbf{M}\mathbf{y})^T (\mathbf{b} - \mathbf{M}\mathbf{y}) \right\} \\ &= \mathbf{E} \left\{ \text{tr} \left[(\mathbf{b} - \mathbf{M}\mathbf{y})(\mathbf{b} - \mathbf{M}\mathbf{y})^T \right] \right\} \\ &= \text{tr} \left\{ \mathbf{E} \left[(\mathbf{b} - \mathbf{M}\mathbf{y})(\mathbf{b} - \mathbf{M}\mathbf{y})^T \right] \right\} \\ &= \text{tr} \{ \text{cor}(\mathbf{b} - \mathbf{M}\mathbf{y}) \} \end{aligned}$$

其中 $\text{cor}(\mathbf{b} - \mathbf{M}\mathbf{y}) = \mathbf{E} \left[(\mathbf{b} - \mathbf{M}\mathbf{y})(\mathbf{b} - \mathbf{M}\mathbf{y})^T \right]$ 是自相关矩阵

在加性噪声与用户信号不相关时有

$$\mathbf{cor}(\mathbf{b} - \mathbf{M}\mathbf{y}) = \mathbf{I} + \mathbf{M}(\mathbf{R}\mathbf{A}^2\mathbf{R} + \sigma^2\mathbf{R})\mathbf{M}^T - \mathbf{A}\mathbf{R}\mathbf{M}^T - \mathbf{M}\mathbf{R}\mathbf{A}$$

其中加性噪声的方差为 σ^2

于是目标函数可写作

$$\begin{aligned} J(\mathbf{M}) &= \text{tr}\{\mathbf{cor}(\mathbf{b} - \mathbf{M}\mathbf{y})\} \\ &= \text{tr}(\mathbf{I}) + \text{tr}\left(\mathbf{M}(\mathbf{R}\mathbf{A}^2\mathbf{R} + \sigma^2\mathbf{R})\mathbf{M}^T\right) - \text{tr}(\mathbf{A}\mathbf{R}\mathbf{M}^T) - \text{tr}(\mathbf{M}\mathbf{R}\mathbf{A}) \end{aligned}$$

利用迹函数的微分公式

$$\frac{\partial \text{tr}(\mathbf{M}^T \mathbf{B})}{\partial \mathbf{M}} = \frac{\partial \text{tr}(\mathbf{B}\mathbf{M}^T)}{\partial \mathbf{M}} = \mathbf{B} \quad \frac{\partial \text{tr}(\mathbf{M}\mathbf{B})}{\partial \mathbf{M}} = \frac{\partial \text{tr}(\mathbf{B}\mathbf{M})}{\partial \mathbf{M}} = \mathbf{B}^T$$

$$\frac{\partial \text{tr}(MDM^T)}{\partial M} = M(D + D^T)$$

因为 $D = RA^2R + \sigma^2R$ 是对称矩阵，所以

$$\frac{\partial J(M)}{\partial M} = 2M(RA^2R + \sigma^2R) - 2AR$$

令其为零，即可得

$$M(RA^2R + \sigma^2R) = AR$$

如果 R 非奇异，可得最优的多用户检测器为

$$M = A(RA^2 + \sigma^2I)^{-1}$$

作业

- **P163: 3、4、5、6**
- **P170: 7、8、9**
- **P177: 2、3、4**